# Deep Learning Assisted Robust DOA Estimation for Small-Sized Microphone Array Using Sound Intensity Features

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***Abstract*—High-precision localization of indoor sound source in hostile environments remains a significant challenge. This paper proposes a robust direction of arrival (DOA) estimation approach for a small-sized microphone array. The method integrates sound intensity features with deep learning, designing a novel masking scheme, the Sound Intensity Phase-Sensitive Mask (IPSM), as the network output. It suppresses incoherent intensity components via time-frequency (T-F) weighting, thereby enhancing the robustness of DOA estimation. Additionally, a generalized sound intensity normalization scheme is also proposed for the weighting of T-F units, effectively suppressing noise and reverberation interference. Simulation results demonstrate that the proposed method effectively improves DOA estimation performance in complex acoustic environments.**

***Keywords—sound source localization, sound intensity, small-sized array, deep learning***

## I. Introduction

Localization using microphone arrays is fundamental to real-time conference systems [1], voice-controlled systems [2], and hearing aids [3], among many others. To date, various classical methods have been widely studied, including generalized cross correlation with phase transform (GCC-PHAT) [4], multiple signal classification (MUSIC) [5], and estimation of signal parameters via rotational invariance techniques (ESPRIT) [6]. However, these methods typically rely on the angular resolution of arrays and usually require a large aperture to capture adequate spatial information [7]. In enclosed indoor environments, localization with small-sized arrays is of greater practical interest, yet traditional methods are often limited under such conditions [8].

Compared to traditional methods, sound intensity-based estimation provides a new perspective for localization with small-sized arrays [9]. Furthermore, because of the limited robustness of sound intensity localization methods in complex acoustic environments, recent studies have attempted to incorporate machine learning techniques for improving performance [10]. However, although machine learning has improved the noise resistance of sound intensity localization methods to a certain extent, existing research still cannot provide sufficient localization accuracy in strong noise and high reverberation environments [11-12]. Moreover, in high-noise environments, the decision boundary of classification networks as employed by the existing sound intensity based localization methods is susceptible to interference, leading to abrupt changes in the classification results that hinder network learning, particularly under conditions of high angular resolution [13]. In contrast, regression networks operate in the continuous azimuth domain, preserving richer information, thereby facilitating stable network training [14-15].

This paper proposes a robust direction of arrival (DOA) estimation method in the continuous azimuth domain that integrates a regression network structure with time-frequency (T-F) masking. The main contributions are twofold: First, we introduce a novel sound intensity based T-F masking method, called the Intensity Phase-Sensitive Mask (IPSM). Unlike conventional T-F masks in the sound pressure domain, IPSM operates in the sound intensity domain and performs phase-sensitive weighting on active sound intensity, thereby enhancing directionally consistent T-F units and improving localization robustness under strong reverberation. Second, a generalized sound intensity normalization method is designed to improve the algorithm's denoise capability. By nonlinearly adjusting the relationship between sound pressure and particle velocity components, this method optimizes the contribution weights of each T-F unit, thereby increasing the weights of low-energy T-F units and suppressing the interference of high-energy units, significantly improving the accuracy in high-noise, high-reverberation environments. Simulation results show that the proposed method outperforms its state-of-the-art counterparts in localization accuracy.

## II. Signal Model

Considering that an equilateral triangular array represents the smallest circular array, this paper employs three omnidirectional microphones $M_i$ $(i = 1,2,3)$ to form a small equilateral triangular array for sound source localization, as shown in Fig. 1(a). The array center is defined as the circle center, with the diameter of the circle denoted as $D$. The angle of incidence of the sound source relative to the positive x-axis is $\theta \in [0, 2\pi)$.

This work adopts a pressure-pressure (p-p) sound intensity measurement method based on closely spaced microphones. Using Euler's formula and finite differences, the particle velocity along each microphone pair in the short-time Fourier transform (STFT) domain is estimated as [16]:

$$\widehat{U}_{pq}(t_k, f_b) = \frac{1}{j2\pi f_b \rho \Delta r}\left[P_p(t_k, f_b) - P_q(t_k, f_b)\right], \quad (1)$$

where $P_p(t_k, f_b)$ and $P_q(t_k, f_b)$ denote respectively the STFT of the sound pressure signals received by microphones $M_p$ and $M_q$ at time $t_k$ and frequency $f_b$. $\widehat{(\cdot)}$ denotes an estimated value, and $\widehat{U}_{pq}$ denotes the particle velocity along the array edge from $M_p$ to $M_q$. Here, $(p,q) \in \{(1,2),(1,3),(2,3)\}$, corresponding to $\widehat{U}_{12}$, $\widehat{U}_{13}$, and $\widehat{U}_{23}$. $\rho$ is the air density, $\Delta r$ is the microphone spacing, and $j = \sqrt{-1}$.

By orthogonally decomposing the particle velocities of the three edges of the array, as shown in Fig. 1(b), the x-axis and y-axis particle velocity components in the STFT domain at the array center can be estimated as:

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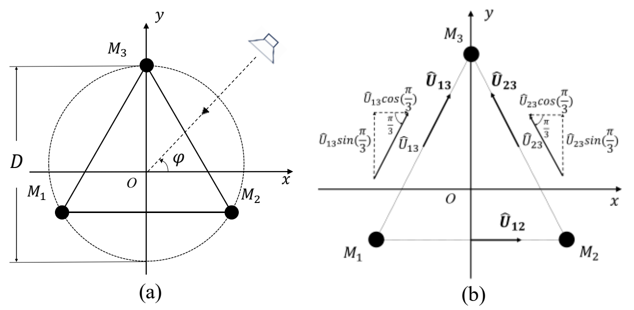


Fig.1. (a) Microphone array configuration and (b) Orthogonal decomposition of particle velocity components

$$\hat{U}_x(t_k,f_b) = \hat{U}_{12}(t_k,f_b) + \frac{1}{2}\hat{U}_{13}(t_k,f_b) - \frac{1}{2}\hat{U}_{23}(t_k,f_b)\,, \tag{2}$$

$$\hat{U}_y(t_k,f_b) = \frac{\sqrt{3}}{2}\hat{U}_{13}(t_k,f_b) + \frac{\sqrt{3}}{2}\hat{U}_{23}(t_k,f_b)\,, \tag{3}$$

where $\hat{U}_x$ and $\hat{U}_y$ denote the particle velocity components along the x-axis and y-axis, respectively.

The sound pressure at the array center is derived by averaging the pressure signals of the three microphones, and its STFT representation $\hat{P}_o(t_k,f_b)$ is given by:

$$\hat{P}_o(t_k,f_b) = [P_1(t_k,f_b) + P_2(t_k,f_b) + P_3(t_k,f_b)]/3\,. \tag{4}$$

Note that only the active sound intensity component of the sound intensity indicates the propagation direction of the sound source. Combining (2)-(4) yields the active sound intensity at the center of the array [17]:

$$\hat{I}_x(t_k,f_b) = \mathrm{Re}\left[\hat{P}_o(t_k,f_b)\hat{U}_x^*(t_k,f_b)\right], \tag{5}$$

$$\hat{I}_y(t_k,f_b) = \mathrm{Re}\left[\hat{P}_o(t_k,f_b)\hat{U}_y^*(t_k,f_b)\right], \tag{6}$$

where $\hat{I}_x$ and $\hat{I}_y$ are the x-axis and y-axis active sound intensity components, $\mathrm{Re}[\cdot]$ denotes the real part, and $(\cdot)^*$ represents the complex conjugate.

Next, summing the sound intensity components across all T-F units to obtain the total x-axis and y-axis intensities. Under the far-field plane wave assumption, the active sound intensity vector indicates the propagation direction. Thus, the azimuth angle of the source $\hat{\varphi}$ can be estimated as [18]:

$$\hat{\varphi} = \arctan\left\{\frac{\sum_{t_k}\sum_{f_b=1}^{N/2}\hat{I}_y(t_k,f_b)}{\sum_{t_k}\sum_{f_b=1}^{N/2}\hat{I}_x(t_k,f_b)}\right\}, \tag{7}$$

where $N$ denotes the number of STFT frequencies units.

## III. Proposed Method

The system diagram of the proposed method is illustrated in Fig. 2(a). The whitening weighting intensity features are first fed into a U-Net network to estimate the IPSM. The estimated mask is then used as an important weighting factor in the post-processing stage, where it is combined with the generalized normalized intensity features and T-F unit selection to achieve DOA estimation. Specifically, these components act as multiplicative weighting terms applied to the active sound intensity, yielding a refined estimate that replaces the direct intensity in (7) for DOA estimation. In the following, we will discuss each module in detail.

### A. Input Features

In [11], an improved whitening weighting method was applied to input feature extraction, thereby enhancing the network's learning capability under high-noise conditions. In contrast, this paper employs a three-element array structure.

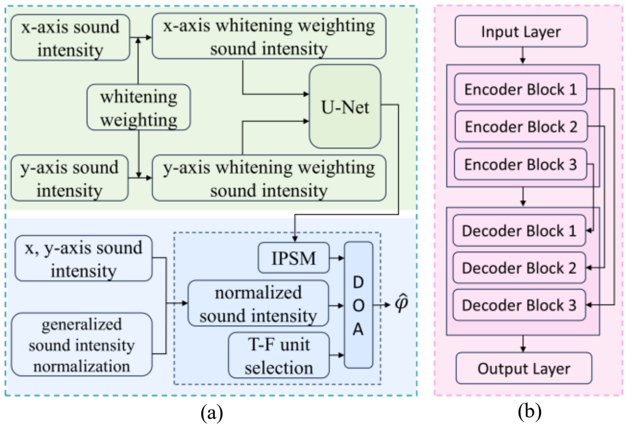


Fig.2. (a) System diagram and (b) Network structure

Compared with the four-element array structure in [11], the number of subarrays is limited. Therefore, only the whitening weighting array-center sound intensity features are used as the network's input. The whitening weighting is defined as:

$$W(t_k,f_b) = \left[\left|\hat{P}_o(t_k,f_b)\right|^2 + \delta\left(\left|\hat{U}_x(t_k,f_b)\right|^2 + \left|\hat{U}_y(t_k,f_b)\right|^2\right)\right]^{\frac{1}{2}}, \tag{8}$$

where δ>0 is a tunable parameter determined during training.

Combining (5), (6), and (8), the whitening weighting sound intensity input features can be expressed as:

$$\hat{I}_x^{(W)}(t_k,f_b) = \frac{1}{W(t_k,f_b)}\mathrm{Re}\left[\hat{P}_o(t_k,f_b)\hat{U}_x^*(t_k,f_b)\right], \tag{9}$$

$$\hat{I}_y^{(W)}(t_k,f_b) = \frac{1}{W(t_k,f_b)}\mathrm{Re}\left[\hat{P}_o(t_k,f_b)\hat{U}_y^*(t_k,f_b)\right]. \tag{10}$$

### B. Sound Intensity Phase-Sensitive Mask

Applying masks to noisy signals is helpful to denoise [19-20]. The commonly-used input masks are the ideal ratio mask (IRM) and the phase-sensitive mask (PSM) [19–21]. Both masks assign T-F weights according to the difference between noisy and clean signals. Specifically, IRM relies on signal energy ratio, while PSM further incorporates the phase difference information.

However, both IRM and PSM are defined in the sound pressure domain, and their weighting mechanisms are not specifically designed for sound intensity. Although these masks perform well in the sound pressure domain, their direct application to intensity estimation is problematic: sound intensity is jointly determined by sound pressure and particle velocity, which prevents the masks from effectively emphasizing the T-F units critical for direction estimation. Motivated by the need for task-consistent features, this paper introduces the concept of PSM into the intensity domain and proposes the Intensity Phase-Sensitive Mask (IPSM). The network-predicted IPSM is applied to the active sound intensity at the array center, to suppress incoherent intensity components via T-F weighting, thus enhancing the robustness of sound intensity-based direction estimation under strong reverberation. The IPSM is defined as:

$$\widehat{IPSM}(t_k,f_b) = max\left\{0, \widehat{IIRM}(t_k,f_b)\cos\left(\angle\hat{I}(t_k,f_b) - \angle\hat{I}_d(t_k,f_b)\right)\right\}, \tag{11}$$

$$\widehat{IIRM}(t_k,f_b) = \sqrt{\frac{\left|\hat{I}_d(t_k,f_b)\right|^2}{\left|\hat{I}_d(t_k,f_b)\right|^2 + \left|\hat{I}(t_k,f_b) - \hat{I}_d(t_k,f_b)\right|^2}} \tag{12}$$

where $\hat{I}(t_k,f_b) = \hat{I}_x(t_k,f_b) + j\hat{I}_y(t_k,f_b)$ denotes the active sound intensity estimated at the array center from the noisy signal, $\hat{I}_d(t_k,f_b)$ from the clean signal.

### C. Generalized Sound Intensity Normalization

Without normalization, a few high-energy but directionally biased T-F units in sound intensity-based DOA estimation can be excessively amplified, reducing the contribution of low-energy yet directionally correct units that have been effectively selected by IPSM. To address this issue, we propose a generalized sound intensity normalization defined as:

$$G(t_k,f_b) = \left[\left|\hat{P}_o(t_k,f_b)\right|^{\xi} + \sigma\left(\left|\hat{U}_x(t_k,f_b)\right|^{\xi} + \left|\hat{U}_y(t_k,f_b)\right|^{\xi}\right)\right]^{\frac{1}{\xi}}, \tag{13}$$

where $G(t_k,f_b)$ denotes the generalized normalized sound intensity, and $\xi \in (0,1)$ controls the nonlinearity of the normalization function, while $\sigma > 0$ scales sound pressure and particle velocity to the same level. With this design, the proposed normalization reduces the weights of high-energy sound intensity T-F units while enhancing that of low-energy units, thereby improving robustness of DOA estimation.

### D. Time-Frequency Unit Selection

The power ratio reflects the variation trend of signal energy within a T-F unit [22]. When the power ratio is very high, it indicates a sudden increase in signal energy within that T-F unit; when the power ratio is close to 1, it indicates that the T-F unit is in a relatively stable state. Under stationary noise conditions, we assume that T-F units with higher power ratios are more likely to contain transient speech components and carry richer effective localization information.

Therefore, we assign higher localization weights to T-F units with high power ratios, while suppressing the contribution of T-F units potentially affected by reverberation and steady-state noise to the localization results. In this paper, we use the power ratio to filter the sound intensity T-F units to further improve the robustness of DOA estimation. The power ratio is defined as:

$$Q(t_k,f_b) = \frac{1}{3}\sum_{i=1}^{3}\frac{|P_i(t_k,f_b)|^2}{|P_i(t_k-1,f_b)|^2}, \tag{14}$$

where $Q(t_k,f_b)$ denotes the average power ratio of the three microphones. We sort all T-F units in $Q(t_k,f_b)$ from largest to smallest and assign weights proportionally, according to:

$$M_Q(t_k,f_b) = \begin{cases} 1 & (t_k,f_b) \in S_1 \\ 0.5 & (t_k,f_b) \in S_2 \backslash S_1 \\ 0 & \text{otherwise} \end{cases}, \tag{15}$$

where $M_Q(t_k,f_b)$ denotes the T-F unit weight determined by power ratio. $S_1$ and $S_2$ denote the sets of the top $q_1$% and $q_2$% T-F units, with the percentages determined empirically. $S_2 \backslash S_1$ denotes the set including $S_2$ but excluding $S_1$.

### E. Network Architecture

The U-Net architecture adopted in this paper is composed of an encoder and a decoder, as shown in Fig. 2(b). The encoder comprises three downsampling stages, each consisting of two $3 \times 3$ convolutional layers with batch normalization and ReLU activation, followed by a $2 \times 2$ max-pooling layer to progressively extract deep features. The encoder channel dimensions are 32, 64, 128, and 256. The decoder performs upsampling via the transposed convolutions and utilizes skip connections to corresponding encoded features, thereby restoring the T-F resolution while preserving more local information [23].

TABLE I. DATASET CONFIGURATION

| | **Training Dataset** | **Validation Dataset** |
|---|---|---|
| **Room Size ($m^3$)** | $9(\pm1) \times 7(\pm1) \times 3(\pm1)$ | $9(\pm0.5) \times 7(\pm0.5) \times 3(\pm0.5)$ |
| **SNR (dB)** | $-5$~$15$ (5 steps) | $-5$~$15$ (5 steps) |
| **$RT_{60}$ (s)** | 0.2 ~1.0 (0.1 steps) | 0.2 ~1.0 (0.1 steps) |
| **Source Distance (m)** | $\{1.25, 1.50, 1.75\}$ | $\{1.25, 1.50, 1.75\}$ |

## IV. EXPERIMENTAL EVALUATIONS

### A. Experimental Setup

The simulated room has a size of $9.64 \times 7.04 \times 2.95$ m$^3$, with the microphone array center located at $(4.82, 3.52, 1.50)$ m. The array size is $D = 0.04$ m, and both the sound source and the array are placed at the same height. Speech signals are randomly selected from the TIMIT database [24], each with a duration of 2.32 s and sampled at 16 kHz. Simulated room impulse responses (RIRs) are obtained using the image source method [25], and non-directional Gaussian white noise is added. The STFT is computed using a Hann window of 512 samples with 50% overlap between consecutive frames. The sound speed is set as 340 m/s, and the distance between the sound source and the array center is 1.5 m.

The room configurations for the training and validation datasets are summarized in Table I. The sound distance is randomly selected from $\{1.25, 1.50, 1.75\}$ m. The signal-to-noise ratio (SNR) ranges from $-5$ to 15 dB with 5 dB steps. The reverberation time ($RT_{60}$) ranges from 0.2 to 1.0 s with 0.1 s steps. During training, both clean speech and noise signals are available, whereas only noisy observations are used during testing. The training, validation, and test sets contain 1372, 212, and 11 mutually exclusive speech signals, from which samples are randomly generated by selecting segments to form source signals.

Considering the strong continuity of regression network's output masks [26], we employ 48-frame units for feature input and output modeling during training and perform segmented prediction on the original 144-frame signal to further reduce the data scale of input features. For all experiments, the U-Net model is trained using the Adam optimizer with an MSE loss function, an initial learning rate of 0.001, a mini-batch size of 128 and a total of 80 epochs. At the 20th epoch, the learning rate is reduced by a decay factor of 0.1 to further facilitate model convergence.

### B. Baseline Methods

To demonstrate the localization performance of the proposed method in challenging environments, three benchmark methods are considered: First, comparisons are made with the state-of-the-art counterparts that also employ sound intensity features [11–12] to assess the performance advantage of the proposed method among sound intensity-based methods. In addition, a non-sound-intensity method, using phase and sound pressure information as features [15], is also compared.

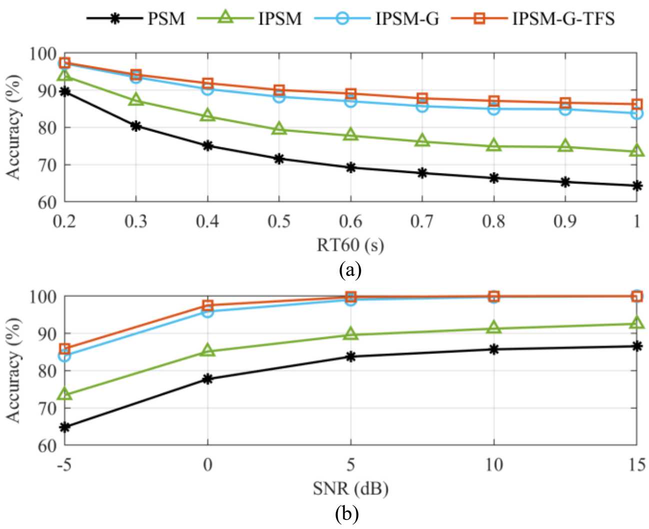


Fig.3. Ablation experimental results: (a) SNR $= -5$ dB and (b) $\mathrm{RT}_{60} = 1.0$ s

All methods are trained and evaluated using the same number of input feature samples for both training and testing. Specifically, the training dataset comprises 400M data units, and the testing dataset contains 100M data units. For the baseline methods, all experimental conditions are kept consistent with those in this paper. Additionally, to improve training efficiency and reduce input data volume, all methods employ a Mel filter bank to compress the input features, consisting of 65 filters [27].

### C. Evaluation Metrics

Localization performance is evaluated over source directions sampled every 1°. We adopt localization accuracy as the performance evaluation metric [28]. Specifically, localization is considered accurate when the absolute error between the estimated angle and the actual azimuth is less than 5°. The localization accuracy rate is defined as:

$$Accuracy = \frac{N_c}{N_s} \times 100\% \,, \tag{16}$$

where $N_c$ denotes the number of correctly localized samples, and $N_s$ denotes the total number of test samples.

### D. Experimental Results

*a) Ablation Experiments:* To more clearly validate the contributions of IPSM, generalized sound intensity normalization, and power ratio T-F unit selection (TFS) to localization performance, we conduct ablation experiments. Specifically, localization experiments are performed using PSM, IPSM, IPSM with generalized sound intensity normalization (IPSM-G), and IPSM-G with TFS (IPSM-G-TFS), respectively, to compare the performance improvements brought by each module.

The parameter $\delta$ in (8) is set to $10^6$, consistent with the value used in [11]. The exponent $\xi$ in (13) is empirically selected under the most challenging condition (SNR $= -5$ dB, $\mathrm{RT}_{60} = 1.0$ s). The optimal value $\xi = 0.6$, corresponding to $\sigma = 24$, is used in all experiments. Empirically, the performance is not sensitive to the exact value of $\xi$ as long as $\xi < 1$. For experiments involving TFS, the threshold parameters $q_1$ and $q_2$ are empirically selected based on validation-set performance, with $q_1 = 35$ and $q_2 = 80$.

TABLE II. DOA ESTIMATION ACCURACY (%) VS. REVERBERATION

| Method | SNR(dB) | $\mathrm{RT}_{60}$(s) | | | | |
|---|---|---|---|---|---|---|
| | | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 |
| Ref. [11] | -5 | 77.26 | 71.06 | 69.51 | 67.53 | 67.11 |
| | 5 | 94.22 | 92.32 | 90.79 | 89.84 | 89.54 |
| Ref. [12] | -5 | 74.27 | 60.31 | 55.24 | 52.41 | 52.27 |
| | 5 | 98.24 | 95.06 | 92.80 | 91.53 | 90.97 |
| Ref. [15] | -5 | 63.97 | 56.81 | 54.26 | 52.43 | 52.43 |
| | 5 | 83.56 | 80.51 | 79.22 | 78.94 | 77.93 |
| Proposed | -5 | 97.41 | 91.89 | 89.11 | 87.14 | 86.26 |
| | 5 | 100.00 | 99.98 | 99.89 | 99.83 | 99.77 |

TABLE III. DOA ESTIMATION ACCURACY (%) VS. SNR

| Method | $\mathrm{RT}_{60}$(s) | SNR(dB) | | | | |
|---|---|---|---|---|---|---|
| | | -5 | 0 | 5 | 10 | 15 |
| Ref. [11] | 0.5 | 70.31 | 85.90 | 91.48 | 93.39 | 93.42 |
| | 1.0 | 66.95 | 83.67 | 89.61 | 91.76 | 91.76 |
| Ref. [12] | 0.5 | 57.27 | 82.84 | 93.66 | 97.44 | 98.42 |
| | 1.0 | 51.77 | 78.04 | 90.51 | 95.65 | 97.19 |
| Ref. [15] | 0.5 | 55.38 | 71.84 | 79.51 | 83.50 | 84.22 |
| | 1.0 | 51.01 | 69.27 | 77.83 | 82.12 | 83.78 |
| Proposed | 0.5 | 90.21 | 98.78 | 99.95 | 100.00 | 100.00 |
| | 1.0 | 86.26 | 97.54 | 99.76 | 99.99 | 100.00 |

As shown in Fig. 3, the proposed IPSM-G-TFS consistently outperforms other methods under both strong noisy conditions with SNR $= -5$ dB (Fig. 3(a)) and strong reverberant conditions with $\mathrm{RT}_{60} = 1.0$ s (Fig. 3(b)). Under the most challenging conditions (SNR $= -5$ dB, $\mathrm{RT}_{60} = 1.0$ s), introducing IPSM, IPSM, generalized sound intensity normalization, and TFS can improve localization accuracy by 9.12%, 10.32%, and 2.46%, respectively. This demonstrates that all modules effectively enhance localization accuracy in highly noisy and reverberant environments.

*b) Effect of Room Reverberation and Additive Noise:* Tables II and III present the localization accuracy comparison between the proposed method and the baseline methods under different $\mathrm{RT}_{60}$ and SNR conditions. All results are averaged over 50 Monte Carlo runs. Table II shows that the proposed method achieves significantly higher localization accuracy than the baseline methods under both SNR conditions. Notably, even in the most adverse noise and reverberation environments, the accuracy remains above 86%, while approaching 100% under moderate SNR conditions. Table III demonstrates that the proposed method also significantly outperforms the baseline methods under two reverberation conditions. Particularly in highly reverberant environments, the accuracy approaches 98% at SNR $= 0$ dB. Under moderate reverberation conditions, it stays close to 100% across nearly all SNR conditions.

## V. CONCLUSION

In this paper, we propose a robust sound intensity-based DOA estimation method for a small-sized microphone array. The method introduces a novel sound intensity based masking scheme, IPSM, to enhance directionally consistent T-F units. In addition, a generalized sound intensity normalization is also proposed for T-F units weighting, to further suppress noise and reverberation. Simulation results show that the proposed method is superior to its state-of-the-art counterparts, especially in highly noise and reverberant environments.

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